

Basics in Statistics

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5/6 October 2026

Who I am

- Lennart
- I live in Bielefeld
- Postdoc in the Econometrics Group at Bielefeld University
- Doctorate in 2026 on modelling heterogeneity in choice behaviour
- Econometrics = economic theory + mathematics + statistics
- No background in biology or medicine, unfortunately

And who are you?

Please raise your hand if you agree: 

- I work in biology
- I work in medicine
- I am in the first or second year of my doctorate
- I have used R before
- I can draw a boxplot
- I would have no trouble explaining to a colleague what a p-value is
- I have run the Münster Marathon, or I plan to

About this course

- The basics of statistics: No fancy methods, just what everyone who wants to extract information from data should know
- Something unclear? Please ask at any time – there are no stupid questions!
- We alternate: I explain something, then you try it out yourselves
- Course materials: loelschlaeger.de/basics_in_statistics
- Solutions to all exercises will be available there after the course

Schedule

Monday, 5 October

- 09 - 12: Statistical Thinking
- 12 - 13: Lunch Break
- 13 - 16: Standard Tools

Tuesday, 6 October

- 09 - 12: Regression Analysis
- 12 - 13: Lunch Break
- 13 - 16: Looking Beyond the Basics

Statistical Thinking

5 October 2026, Morning

Outline

Why statistics?

- Four motivating examples

- What is statistics?

- Deterministic and stochastic processes

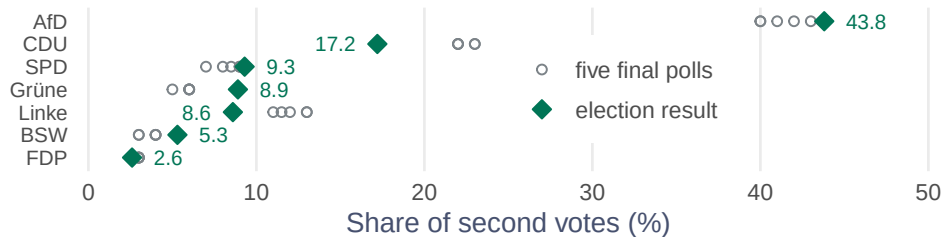
- Populations and samples

- Variables and measurement scales

How do we describe and visualize data?

Probability and distributions

Example 1: Election polls



State election in Saxony-Anhalt, 6 September 2026 (polls and final result)

Exercise 1.1 (3 minutes, in pairs)

1. What is a *sample*, what is a *population*? What are they in this example?
2. Name reasons why a poll can deviate from the election result.

Example 2: Overbooking

An aircraft has 180 seats. Experience shows that some ticket holders do not show up, so airlines sell more tickets than seats.

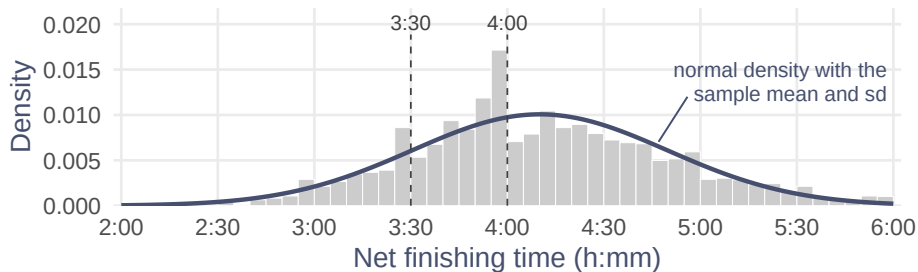
Assumptions: Every ticket holder shows up with probability 95%, independently of the others. A ticket costs 150 EUR. A passenger who is denied boarding receives 400 EUR.

Exercise 1.2 (3 minutes, in pairs)

The airline sells 181 tickets for the 180 seats.

1. What is the probability that all 181 ticket holders show up?
2. What is the probability that every passenger who shows up gets a seat?
3. Difficult: How many tickets should the airline sell to maximize its profit?

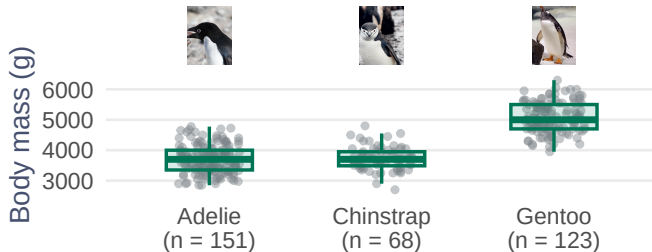
Example 3: Finishing times of the Münster Marathon



Net finishing times of the 3370 finishers, 13 September 2026

- What is a *density*?
- Do the finishing times follow a normal distribution?

Example 4: Body mass of three penguin species



Body mass of 342 penguins of three species

```
install.packages(c("palmerpenguins", "dplyr", "ggplot2"))
library(palmerpenguins)
library(ggplot2)
head(penguins)
ggplot(penguins, aes(x = species, y = body_mass_g)) +
  geom_boxplot()
```

R Quickstart

- RStudio has four panes: The **script** (top left), the **console** (bottom left), the **environment** (top right), and **plots** and help (bottom right).
- Open `u1_statistical_thinking.R` from the course page.
- `Cmd+Enter` (Mac) or `Ctrl+Enter` (Windows) runs the line under the cursor, and its output appears in the console.
- If a package is missing, remove the `#` in front of `install.packages()` at the top of the script and run that line once.

```
library(palmerpenguins) # load the penguin data
?mean                  # help page of a function
x <- penguins$body_mass_g
mean(x, na.rm = TRUE)  # ignore the missing values
```

Discussion

What is statistics?

What are the basics in statistics?

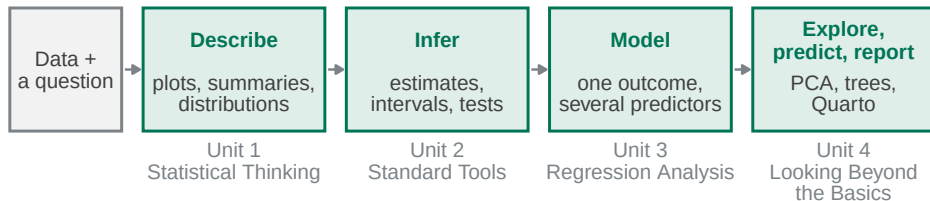
What do you want to be more comfortable with?

Three minutes in pairs, then we collect your answers.

What is statistics?

Definition (Statistics)

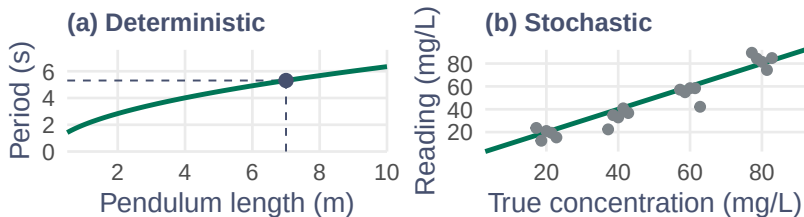
Statistics is the science of learning from data: Collecting data, describing them, and drawing conclusions about a population or a process from a sample, together with a statement of the uncertainty of these conclusions.



Deterministic and stochastic processes

Definition (Deterministic and stochastic)

A process is **deterministic** if the same input always gives the same output. It is **stochastic** if the output varies although the input is the same.



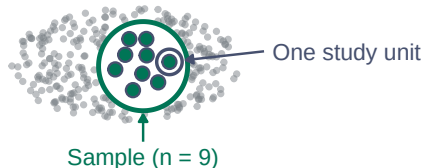
Statistics deals with stochastic processes such as the readings in (b).

Population and sample

Definition (Study unit, population, sample)

A **study unit** is the object on which a measurement is taken. The **population** is the set of all study units the question is about. The **sample** is the n study units actually observed.

Population



Example (Election poll)

Study unit: One eligible voter. Population: All eligible voters of Saxony-Anhalt.
Sample: The 1044 respondents of a poll.

Variables and measurement scales

Definition (Variable, observation)

A **variable** X is a property measured on every study unit; its value on study unit i is the **observation** x_i .

Definition (Types of variables and measurement scales)

categorical	nominal	unordered categories	species	mode
	ordinal	ordered, no distances	tumour grade (I–IV)	median
numerical	interval	differences, no true zero	temperature (°C)	mean
	ratio	differences and ratios	body mass	mean

Interval and ratio scales together are called **metric**. A numerical variable is **discrete** if it takes countably many values (counts) and **continuous** if it can take every value in an interval (measurements). The last column gives the usual measure of centre.

Exercise: Variables and measurement scales

Exercise 1.3 (3 minutes, in pairs)

The data of a marathon contain, for every finisher:

country T-shirt size (S, M, L, XL) year of birth
number of marathons finished before net time

1. Give the type (categorical or numerical, discrete or continuous) and the measurement scale of each of the five variables.
2. For which of the variables is a mean meaningful?

Outline

Why statistics?

How do we describe and visualize data?

- Measures of centre

- Measures of spread

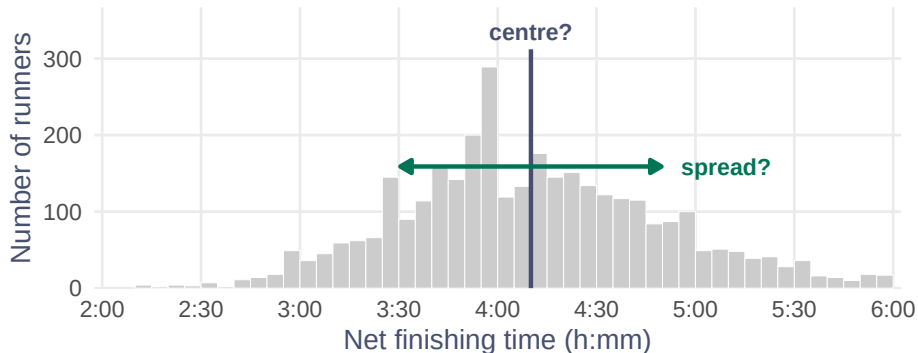
- Shape, boxplot, and outliers

- Graphical displays

- Describing and visualizing data in R

Probability and distributions

Describing a variable



Net finishing times of the 3370 finishers of the Münster Marathon 2026

- Where is the **centre** of the values, and how do we measure it?
- How much do the values **spread** around that centre?

Mean, median, mode

Definition (Measures of centre)

For observations $x_1, \dots, x_n$:

- the **arithmetic mean** is $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$;
- the **median** x_{med} divides the sorted observations into two halves: It is the middle value (n odd) or the mean of the two middle values (n even);
- the **mode** is the most frequent value.

Example (Münster Marathon, 3370 net times)

Mean 4:10, median 4:07. Women: Mean 4:21, median 4:19 ($n = 766$). Men: Mean 4:07, median 4:01 ($n = 2604$).

Variance, standard deviation, quantiles

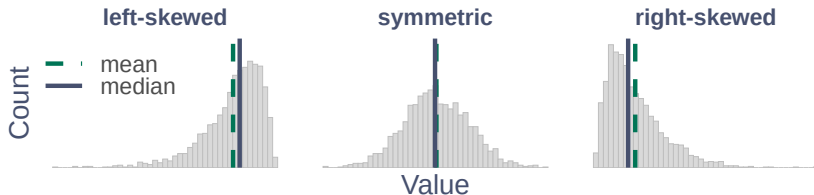
Definition (Measures of spread)

- **Sample variance** and **standard deviation**: $s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$, $s = \sqrt{s^2}$.
- The **p -quantile** x_p is the value below which a share p of the sorted observations lies: A quarter of the runners are faster than $x_{0.25}$. Quartiles: $x_{0.25}$, $x_{0.5}$ (the median), $x_{0.75}$.
- **Interquartile range**: $\text{IQR} = x_{0.75} - x_{0.25}$. **Range**: $x_{\max} - x_{\min}$.

Example (Münster Marathon)

$s = 39.6$ minutes. Quartiles 3:44 and 4:36, so the middle half of the field finishes within $\text{IQR} = 52.6$ minutes. Range 2:10 to 5:59.

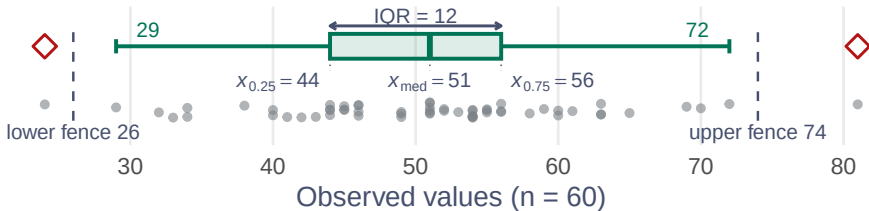
Symmetric and skewed distributions



Definition (Skewness)

A distribution is **right-skewed** if its right tail is longer; then typically $\text{mode} < \text{median} < \text{mean}$. It is **left-skewed** if the left tail is longer. A distribution with one peak is **unimodal**, with two peaks **bimodal**.

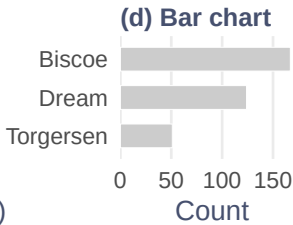
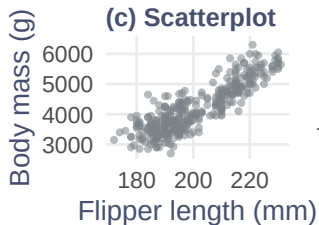
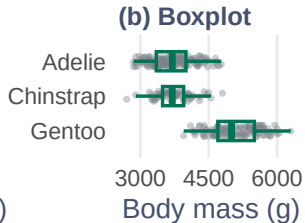
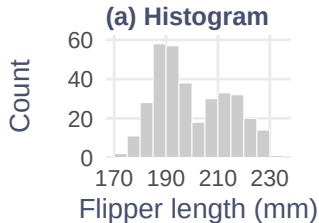
Boxplot and outliers



Definition (Boxplot)

The box runs from $x_{0.25}$ to $x_{0.75}$ with a line at the median. The whiskers end at the most extreme observations within the fences $x_{0.25} - 1.5 \text{ IQR}$ and $x_{0.75} + 1.5 \text{ IQR}$.

Standard displays



histogram

distribution of one numerical variable

boxplot

comparison of a numerical variable between groups

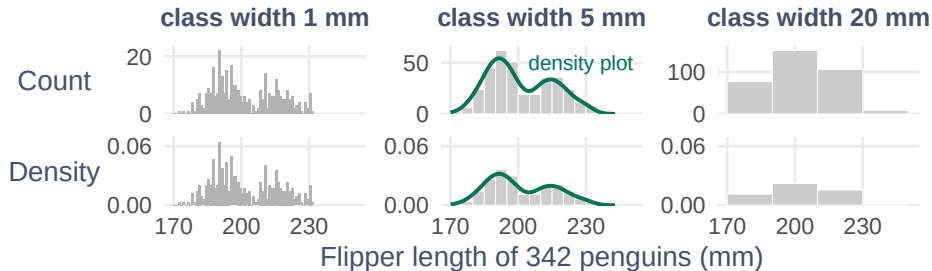
scatterplot

relationship between two numerical variables

bar chart

frequencies of a categorical variable

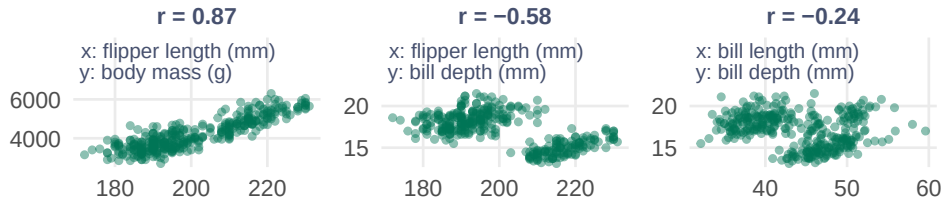
Histograms and class width



Definition (Histogram)

The range is divided into classes. Over each class a bar is drawn whose *area* is proportional to the number of observations in the class. On the **count scale** the height is the count (equal class widths). On the **density scale** the height is the relative frequency divided by the class width, so that the total area is 1.

Correlation



Definition (Correlation coefficient)

For pairs $(x_1, y_1), \dots, (x_n, y_n)$ the **Pearson correlation**

$$r = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i (x_i - \bar{x})^2 \sum_i (y_i - \bar{y})^2}},$$

with $-1 \leq r \leq 1$, measures direction and strength of the *linear* association.

Exercise: Describing and visualizing data by group

Go to `u1_statistical_thinking.R`, section 1.1

R exercise 1.1 (15 minutes, in pairs)

1. Work through section 1.1: Run every demo, then solve the task below it.
2. Which species is the heaviest, and which two are hard to tell apart?
3. Is the heaviest species also the one with the longest flippers?
4. Which measurement goes with body mass most closely?

Outline

Why statistics?

How do we describe and visualize data?

Probability and distributions

- What is a probability?

- Rules of probability

- Conditional probability

- Diagnostic tests

- Random variables

- Discrete distributions

- The normal distribution

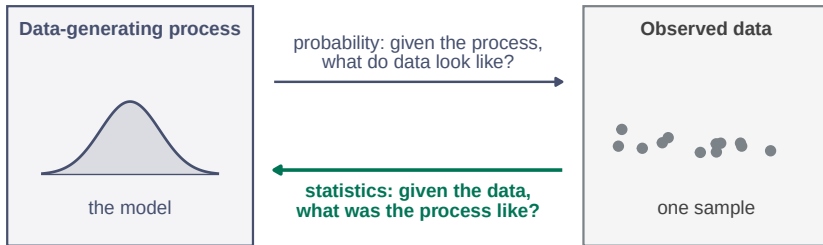
Why probability?

Data are subject to random variation.

Statistics therefore has to answer questions such as:

- Is an observed difference systematic or due to chance?
- How precise is an estimate?
- How can decisions be made under uncertainty?

Probability and statistics



Definition (Statistical model)

A **statistical model** describes the **data-generating process**: Which values are possible and with which probabilities. The observed data are one realization of this process.

Exercise: Five of a kind

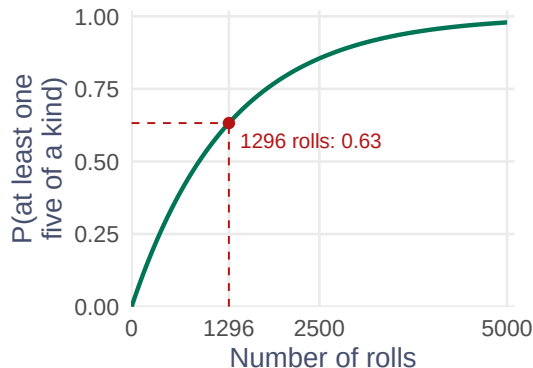
A man rolls five dice. The probability that all five show the same number is 1 in 1296. So he rolls them 1296 times, and five of a kind never comes up. His verdict: Science is wrong.

Video: [youtube.com/shorts/00kz3SDr_Yg](https://www.youtube.com/shorts/00kz3SDr_Yg)

Exercise 1.4 (3 minutes, in pairs)

1. Where does 1 in 1296 come from?
2. Did he prove science wrong?

Misconception of probability



Independent rolls: No five of a kind in 1296 rolls has probability

$$\left(\frac{1295}{1296}\right)^{1296} = 0.37.$$

Complement:

$$P(\text{at least one}) = 0.63.$$

Exercise: Red is due

European roulette on a fair wheel has 37 pockets: 18 red, 18 black, and a green zero. At a table, black has come up 10 times in a row. Many players now bet on red: “Red is due.”

Exercise 1.5 (3 minutes, in pairs)

1. What is the probability of red on a single spin?
2. After 10 times black: Is red now more likely than black?

What is a probability?

A **probability** is a number between 0 (impossible) and 1 (certain) that states how likely an event A is. $P(A) = 0.25$ means 25 %.

Equally likely outcomes

$P(A)$ is the share of the outcomes that belong to A .

One die: $P(\text{six}) = 1/6$.

Two coins (H = heads, T = tails):

	H	T	
H	HH	HT	$P(HH) = 1/4$ $P(HT \text{ or } TH) = 2/4$
T	TH	TT	

Long-run relative frequency

$P(A)$ is the share of A in a long series of independent repetitions.

Airline: 95 % of ticket holders show up.

Seed lot: 80 % of the seeds germinate.

Prevalence: 1 % of the women in a region have breast cancer.

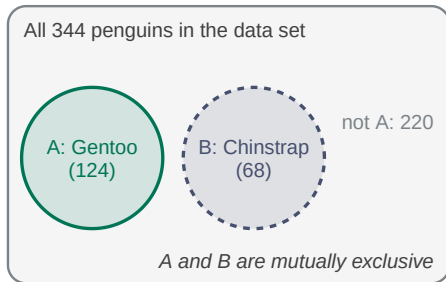
Random experiments and events

Definition (Random experiment, event)

A **random experiment** is a process with an uncertain outcome. An **event** is a statement about the outcome that is either true or false.

random experiment	event
test one patient	the test is positive
sow one seed	the seed germinates
draw one penguin	the bird is a Gentoo

Events can be combined: Not A (the **complement**), A and B , A or B . **Mutually exclusive** events cannot occur together.



Independent events

Definition (Independence, multiplication rule)

Two events are **independent** if the occurrence of one does not change the probability of the other. Then $P(A \text{ and } B) = P(A) \cdot P(B)$.

		First die					
		1	2	3	4	5	6
Second die	1						
	2						
	3						
	4						
	5						
	6						

Two dice: $P(\text{two sixes}) = 1/6 \cdot 1/6 = 1/36$, one of 36 equally likely cells.

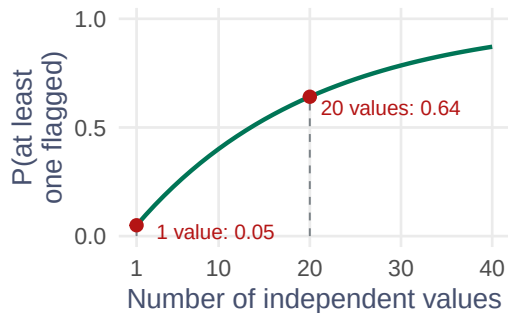
- Roulette:
 $P(10 \text{ times black}) = (18/37)^{10} = 0.00074$
- All 181 ticket holders show up: $0.95^{181} = 0.0001$

Independence fails for repeated measurements on the same animal or for patients from the same family.

The complement rule

Definition (Complement rule)

$P(\text{not } A) = 1 - P(A)$. Most useful for “at least one”: $P(\text{at least one}) = 1 - P(\text{none})$.



A laboratory flags a value outside the middle 95 % of healthy people. For 20 independent values of a healthy person, $P(\text{none flagged}) = 0.95^{20} = 0.36$, so $P(\text{at least one}) = 0.64$.

Five of a kind in 1296 rolls:
 $1 - (1295/1296)^{1296} = 0.63$.

The same calculation applies to running 20 statistical tests at the 5 % level (Unit 2).

Exercise: Rules of probability

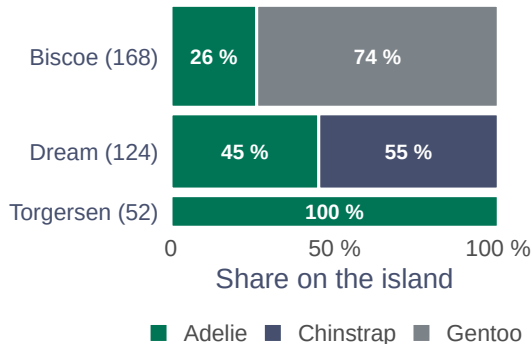
Exercise 1.6 (5 minutes, in pairs)

1. Each seed of a lot germinates with probability 0.8, independently of the others. Three seeds are sown. What is the probability that all three germinate, that none germinates, and that at least one germinates?
2. You guess the answer to a multiple-choice question with four options. What is the probability that your guess is correct? That at least one of three guesses is correct?
3. The three seeds are sown in the same pot. Why is independence questionable?

Conditional probability

Definition (Conditional probability)

The **conditional probability** $P(A | B)$, read “ A given B ”, is the probability of A when it is known that B has occurred. It is the share of A among the outcomes in B :
$$P(A | B) = P(A \text{ and } B) / P(B).$$



Penguins: $P(\text{Gentoo}) = 0.36$, but
 $P(\text{Gentoo} | \text{Biscoe}) = 0.74$.

Screening: $P(\text{positive} | \text{disease})$ and
 $P(\text{disease} | \text{positive})$ can differ greatly.

A and B are independent exactly if
 $P(A | B) = P(A)$.

Example: Conditional probability

The penguin data: 344 birds of three species (rows: Adelie, Chinstrap, Gentoo), each recorded on one of three islands of the Palmer Archipelago in Antarctica (columns: Biscoe, Dream, Torgersen).

species	island			Sum
	Biscoe	Dream	Torgersen	
Adelie	44	56	52	152
Chinstrap	0	68	0	68
Gentoo	124	0	0	124
Sum	168	124	52	344

Draw one of the 344 penguins at random.

$$P(\text{Gentoo}) = 124/344 = 0.36$$

$$P(\text{Gentoo} \mid \text{Biscoe}) = 124/168 = 0.74:$$

74 % of the birds on Biscoe are Gentoo.

$$P(\text{Biscoe} \mid \text{Gentoo}) = 124/124 = 1:$$

Every Gentoo was recorded on Biscoe.

A conditional probability is a row or column share of the table.

Exercise: Conditional probability

species	island			Sum
	Biscoe	Dream	Torgersen	
Adelie	44	56	52	152
Chinstrap	0	68	0	68
Gentoo	124	0	0	124
Sum	168	124	52	344

Exercise 1.7 (4 minutes, in pairs)

Draw one of the 344 penguins at random.

1. Compute $P(\text{Adelie} \mid \text{Dream})$ and $P(\text{Dream} \mid \text{Adelie})$. What do they mean?
2. Compare $P(\text{Adelie} \mid \text{Dream})$ with $P(\text{Adelie})$. Are “Adelie” and “Dream” independent?

Sensitivity, specificity, and predictive values

A diagnostic test is described by conditional probabilities:

		share of ...
prevalence	$P(\text{disease})$	the diseased among all tested
sensitivity	$P(\text{positive} \mid \text{disease})$	the positives among the diseased
specificity	$P(\text{negative} \mid \text{healthy})$	the negatives among the healthy
false-positive rate	$P(\text{positive} \mid \text{healthy})$	the positives among the healthy
positive predictive value	$P(\text{disease} \mid \text{positive})$	the diseased among the positives
negative predictive value	$P(\text{healthy} \mid \text{negative})$	the healthy among the negatives

Sensitivity and specificity describe the test. The positive predictive value, which a patient wants to know, also depends on the prevalence.

Example: Mammography screening

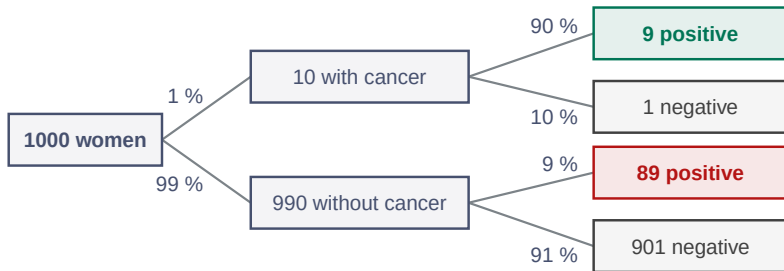
Breast cancer screening by mammography in a region:

- 1 % of the women have breast cancer (prevalence)
- 90 % of the women with cancer test positive (sensitivity)
- 9 % of the women without cancer test positive (false-positive rate)

A woman's mammogram is positive. She asks whether she has breast cancer. What is the best answer?

- (A) The probability that she has breast cancer is about 81 %.
- (B) About 9 of 10 women with a positive mammogram have breast cancer.
- (C) About 1 of 10 women with a positive mammogram has breast cancer.
- (D) The probability that she has breast cancer is about 1 %.

Natural frequencies



Of the $9 + 89 = 98$ women with a positive mammogram, only 9 have breast cancer: $P(\text{cancer} \mid \text{positive}) = 9/98 = 0.09$. Answer (C).

In 2007, only 21 % of 160 gynaecologists chose (C); after a short training with natural frequencies, 87 % did (Gigerenzer et al., 2007). The computation behind the tree is known as **Bayes' theorem**.

Exercise: A positive pregnancy test

Exercise 1.8 (5 minutes, in pairs)

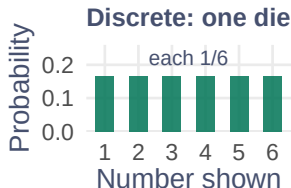
A pregnancy test has a sensitivity of 99 % and a false-positive rate of 2 % (assumed numbers).

1. Before an X-ray examination, 10 000 women are tested routinely, and 2 % of them are pregnant. Draw the tree: How many are pregnant, how many of them test positive, and how many of the others test positive? What is the positive predictive value?
2. A woman takes the test because her period is late. Assume that half of the women in this situation are pregnant. What is the positive predictive value now?
3. The test is the same in 1 and 2. Why are the answers so different?

Random variables

Definition (Random variable, distribution)

A **random variable** X is a numerical outcome of a random experiment. Its **distribution** states which values are possible and how probable they are.



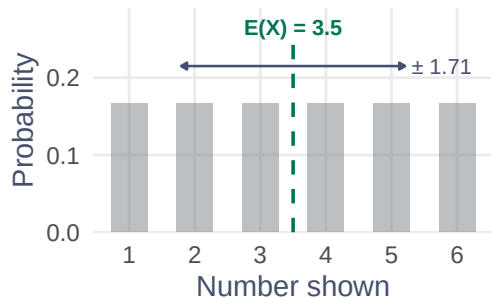
Discrete: Counts. Every value has a probability, drawn as a bar. The bars add up to 1.

Continuous: Measurements. Probabilities are areas under the **density** curve.

Mean and standard deviation of a distribution

Like a data set, a distribution has a centre and a spread:

- The **expected value** $\mu = E(X)$ is the long-run average of X .
- The **standard deviation** σ is the typical distance of X from μ .



$\bar{x}$ and s from the data estimate μ and σ .

One die:

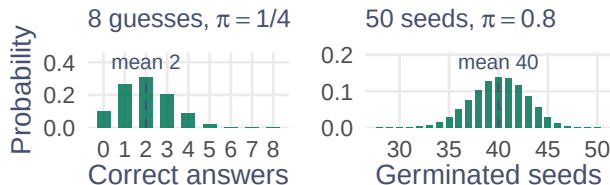
$E(X) = (1 + 2 + \dots + 6)/6 = 3.5$, a value the die never shows.

	data	distribution
mean	$\bar{x}$	μ
standard deviation	s	σ
variance	s^2	σ^2

Binomial distribution

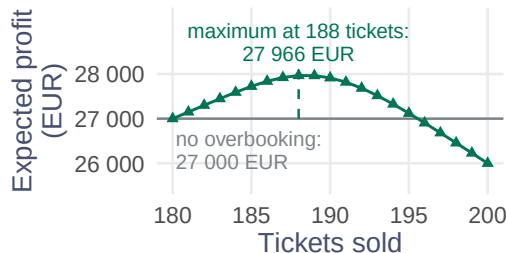
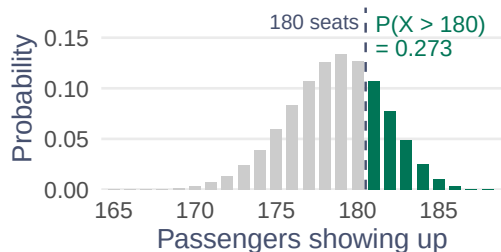
Definition (Binomial distribution)

X is the number of successes in n independent trials, each with the same success probability π : $X \sim \text{Bin}(n, \pi)$, with mean $n\pi$ and variance $n\pi(1 - \pi)$. A single yes/no outcome ($n = 1$) has a **Bernoulli distribution**.



- Guessing all 8 questions of a multiple-choice test with four options each: On average 2 correct answers, none correct with probability $0.75^8 = 0.10$.
- In R: `dbinom(x, n, p)` is $P(X = x)$, `pbinom(x, n, p)` is $P(X \leq x)$.

Example: Overbooking



Example (188 tickets for 180 seats)

Passengers showing up: $X \sim \text{Bin}(188, 0.95)$ with $E(X) = 178.6$ and $\sigma = 2.99$.

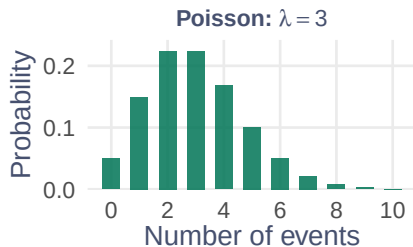
$P(X > 180) = 1 - P(X \leq 180) = 0.273$, in R `1 - pbinom(180, 188, 0.95)`.

Right: Expected profit $150n - 400E[\max(X - 180, 0)]$ in EUR for n tickets sold, with its maximum of 27 966 EUR at $n = 188$, against 27 000 EUR without overbooking.

Poisson distribution

Definition (Poisson distribution)

X counts events in a fixed interval of time or space, when the events occur independently at a constant rate: $X \sim \text{Po}(\lambda)$, with mean and variance both equal to λ .

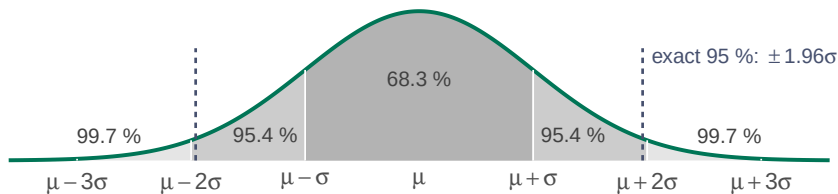


- Examples: Colonies on an agar plate, cells in one square of a counting chamber, new cases of a rare disease per year.
- Real counts often vary more than the model allows (**overdispersion**), for example accidents per driver, because some drivers are more careful than others. The negative binomial distribution allows for this.

Normal distribution

Definition (Normal distribution)

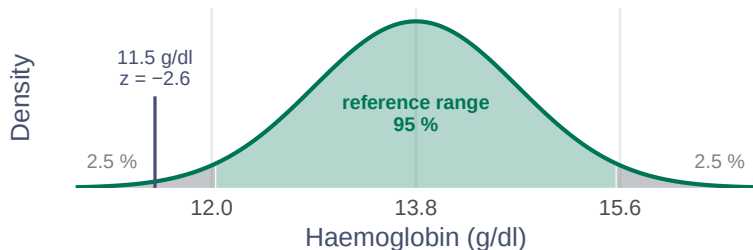
The **normal distribution** $\mathcal{N}(\mu, \sigma^2)$, or Gaussian, has a symmetric, bell-shaped density with mean μ and standard deviation σ . Many measurements are approximately normal, for example body height, or repeated measurements of one sample in the laboratory.



Probability within one, two, and three standard deviations of the mean. In R: `pnorm(x, mean, sd)` is $P(X \leq x)$, with the standard deviation, not the variance.

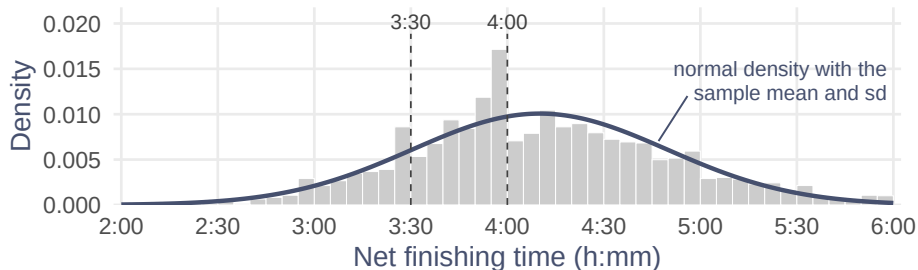
Example: Reference ranges

Haemoglobin of healthy women is approximately normal with $\mu = 13.8$ and $\sigma = 0.9$ g/dl (assumed numbers).



- The **reference range** holds the middle 95 % of healthy people: $\mu \pm 1.96 \sigma$, from 12.0 to 15.6 g/dl. By construction, 5 % of the healthy lie outside.
- The **z-score** $z = (x - \mu)/\sigma$ is the distance from the mean in standard deviations. 11.5 g/dl gives $z = -2.6$: Only 0.5 % of healthy women lie this low.

Example: Are the finishing times normally distributed?



The 3370 net finishing times in 5-minute classes and the normal density with mean 4:10 and standard deviation 39.6 minutes

- The curve fits roughly: One peak, most times between 3:00 and 5:30.
- But the classes just below 3:30 and 4:00 are fuller than the curve predicts, and no time exceeds 6:00.

Exercise: Normal distribution

Exercise 1.9 (8 minutes, in pairs)

Assume that the net finishing time T in minutes follows $\mathcal{N}(250, 40^2)$.

1. The z-score of 240 by hand, then $P(T < 240)$ in R: `pnorm(240, 250, 40)`.
2. Between which two times do the middle 95 % of the finishers lie under this model?
3. Observed: 45.1 % of the 3370 finishers needed less than 4:00, and nobody needed more than 6:00. Compare with the model.
4. Name a natural first model (Bernoulli, binomial, Poisson, or normal) and its parameters: (a) Whether one patient responds to a treatment; (b) the number of seeds out of 50 that germinate; (c) bacterial colonies on an agar plate; (d) new cases of a rare disease in Münster per year; (e) the body height of adult women.

Exercise: Binomial, Poisson, and normal in R

Go to `u1_statistical_thinking.R`, section 1.2

R exercise 1.2 (15 minutes, in pairs)

1. Work through section 1.2: Run every demo, then solve the task below it.
2. 50 seeds germinate independently with probability 0.8. How likely are at least 45?
3. A rare disease has 2 new cases per year on average. How likely are 5 or more?
4. Draw the distribution of the germinated seeds and the histogram of flipper length with a normal curve on top: Does the normal distribution fit?